

Unit-I

Numerical techniques

**Numerical solution of algebraic and transcendental
equations:**

Bisection methods



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Algebraic Equation:

An equation $f(x) = 0$ is called n^{th} degree algebraic equation.

Where $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_n$, real number, $a_0 \neq 0$ and n is non-negative integer.

Transcendental Equation:

An equation which contains trigonometric function, logarithmic function, exponential function etc, is called transcendental equation.

Ex: (i) $\tan x - e^x = 0$.

(ii) $x + \log_e x = 0$.

Nonlinear Equations

- Nonlinear equations involve polynomial of degree greater than one and/or functions like exponential, trigonometric, logarithmic etc.
- Equations that cannot be solved by analytical/direct methods such as

(i) $\sin x - x = 0$.

(ii) $\tan x - e^x = 0$.

(iii) $x + \log_e x = 0$.

Root:

The value α of x which satisfies $f(x) = 0$, is called a root of $f(x) = 0$.

Bisection Method

Bisection method is the simplest **iterative method** to solve nonlinear equations

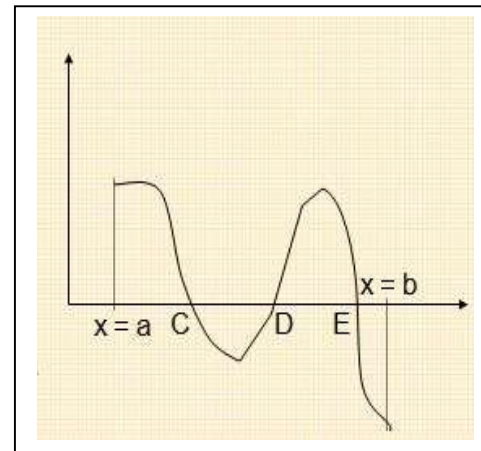
Note: If $f(x)$ is **continuous** and **real** in the interval $[a, b]$ and $f(a), f(b)$ are of **opposite signs**, then the equation $f(x)$ will have atleast one real root between a and b . **e.g.** $f(x) = x^3 - 5$

$$f(0) = -5,$$

$$f(1) = -4,$$

$$f(2) = 3.$$

There lies a root in the **interval (1, 2)**.

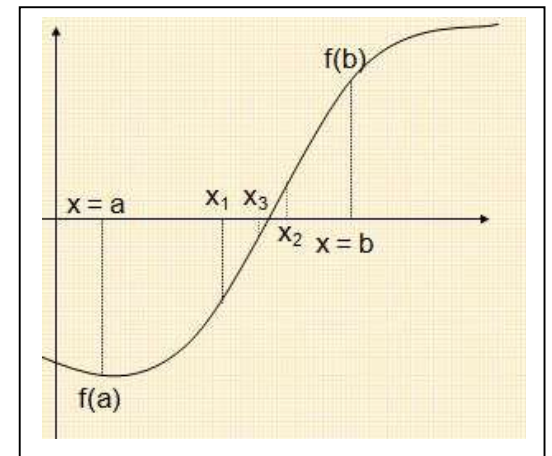


Procedure/Algorithm

- Let the function $f(x)$ be continuous in the interval $[a, b]$.
- Decide initial values of a and b that would **bracket** the root.
- Calculate $f(a)$ and $f(b)$.
- If $f(a).f(b) > 0$, no root exist between a and b .
- Now if $f(a).f(b) < 0$,

Calculate $x_1 = \frac{a+b}{2}$, and hence find $f(x_1)$.

- If $f(x_1) = 0$, **then x_1 is the root.**
- If $f(x_1).f(b) < 0$,



- Root not lies between a and x_1 .
- Root lies between x_1 and b .
- Repeat the procedure till the desired result is obtained.

Example: Solve the equation $x^3 - 9x + 1 = 0$ for the root lying between 2 and 3. Answer should be correct to two decimal places.

Solution: We have

$$\begin{aligned}f(x) &= x^3 - 9x + 1, \\f(2) &= -9, f(3) = 1 \\ \Rightarrow f(2)f(3) &< 0.\end{aligned}$$

Therefore, a root of function $f(x)$ lies between 2 and 3.

Let $a_0 = 2, b_0 = 3 \Rightarrow x_1 = \frac{2+3}{2} = 2.5$.

Now, $f(x_1) = -5.8$.

Hence, a root of function $f(x)$ lies between 2.5 and 3.

Repeating the above process, we have

n	a_n	b_n	$x_{n+1} = \frac{a_n + b_n}{2}$	$f(x_{n+1})$
0	2	3	2.5	-5.8
1	2.5	3	2.75	-2.9
2	2.75	3	2.88	-1.03
3	2.88	3	2.94	-0.05
4	2.94	3	2.97	0.47
5	2.94	2.97	2.955	0.21

6	2.94	2.955	2.9475	0.08
7	2.94	2.9475	2.9438	0.017
8	2.94	2.9438	2.9419	-0.016
9	2.9419	2.9438	2.9428	0.03

In the 8th step a_n, b_n and x_{n+1} are equal up to **two decimal places**. We can take 2.94 as a root up to two decimal places. Therefore the root of $x^3 - 9x + 1 = 0$ is 2.94.

Q: 1. Solve the equation $x^3 - 4x - 9 = 0$ using bisection method such that the root lying between 2 and 3. Answer should be correct to 3 decimal places. **[Answer: $x=2.706$]**

Q.2: Using the bisection method, find an approximate root of the equation $x \sin x = 1$ that lies between 1 and 1.5. Carry out computation upto 7th stage. **[Answer: $x=1.11328$]**

Q.3: Find the positive root of $\cos(x) - xe^x = 0$ using Bisection method.
[Answer: $x=0.5156$]

Thank You